UM2460

Low Power 2.4 GHz Transceiver

Design Specification

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Low Power 2.4 GHz Transceiver

Introduction

The UM2460 integrates a wireless RF transceiver operating at 2.4 GHz, baseband and MAC layer function blocks. The UM2460 can be combined with a microprocessor (e.g. 8051) for low data rate applications such as wireless keyboard, wireless mouse, home automation, consumer electronics, PC peripherals, toys, industrial automation, etc. and other medium data rate applications like wireless voice and image links. The UM2460 uses advanced radio architecture to minimize the external component count and the power consumption. The RF block of the UM2460 integrates receiver, transmitter, VCO and PLL within a single IC. The UM2460 MAC/Baseband mainly consists of TX/RX FIFOs, receiving frame filter and digital signal processing module. The UM2460 is fabricated with advanced 0.18µm CMOS process.

Features

RF/Analog

- ISM band 2.405~2.480 GHz operation
- -90 dBm sensitivity @250Kbps
  -80 dBm sensitivity @1Mbps
  (Packet error rate under 0.1%)
- -3 ~ 0 dBm typical output power
- Integrated 32 MHz oscillator drive
- PLL lock-on time less than 130 us
- Low current consumption, 18 mA in RX and 15 mA in TX mode, under 1Mbps.
- 2.5 uA deep sleep mode
  <1 uA power down mode
- 0.18 µm CMOS technology

MAC/Baseband

- Automatic ACK response and FCS check
- ACK packet includes address information (4 bytes) and 2-byte user information
- TX FIFO – 64 bytes

- Dual RX FIFOs – 64 bytes each

Table of Content

1. Pin Configuration

Device Pin Descriptions

Pin type abbreviation: A = Analog, D = Digital, I = Input, O = Output, P = Power

Pin	Symbol	Type	Description
1	DB5	Ground	DB5
2	VDD_RF1	Power	RF power supply. Bypass with a capacitor as close to the pin as possible.
3	DB2	Ground	DB2
4	RF_P	AIO	Differential RF input/output (+)
5	RF_N	AIO	Differential RF input/output (-)
6	DB2	Ground	DB2
7	VDD_RF2	Power	RF power supply. Bypass with a capacitor as close to the pin as possible.
8	SCAN_MODE	DI	Digital scan/test mode enable signal for chip testing purpose
9	SCAN_EN	DI	Scan insertion testing enable signal
14	INT	DO	Interrupt pin to the micro-processor
15	RESETn	DI	Global hardware reset pin, active low
16	WAKE	DI	External wake up trigger
17	SO	DO	Serial interface data output
18	SI	DI	Serial interface data input
19	SCLK	DI	Serial interface clock
20	SEN	DI	Serial interface enable
21	GPIO2	DO	General purpose digital I/O, also used as an external TX/RX switch control
22	GPIO1	DO	General purpose digital I/O; also used as an external TX/RX switch control
23	GPIO0	DO	General purpose digital I/O; also used as an external PA enable
24	VDD_D	Power	Digital circuit power supply
25	GND_D	Ground	Digital ground.
26	VDD_3V	Power	3V input for a DC-DC regulator
27	VDD_2V2	AO	DC-DC output voltage
28	GND_GR	Ground	Guad-Ring ground.
29	VDD_GR	Power	Guard ring power supply. Bypass with a capacitor as close to the pin as possible.
30	VDD_BG	Power	Power supply for bandgap reference circuit. Bypass with a capacitor as close to the pin as possible.
31	VDD_A	Power	Power supply for an analog circuit. Bypass with a capacitor as close to the pin as possible.
32	DB4	Ground	DB4
33	XTAL_N	AI	32 MHz Crystal input (-)
34	XTAL_P	AI	32 MHz Crystal input (+)
35	VDD_PLL	Power	PLL power supply. Bypass with a capacitor as close to the pin as possible.
36	GND_PLL	Ground	PLL ground.
37	VDD_CP	Power	Charge pump power supply. Bypass with a capacitor as close to the pin as possible.
38	VDD_VCO	Power	VCO supply. Bypass with a capacitor as close to the pin as possible.

39	DB5	Ground	DB5
40	LOOP_C	AIO	PLL loop filter external capacitor. Connected to the external 100 pF capacitor.
41	IC ground pad	Ground	Backside ground plane. Must be connected to the ground.

Table 1 Pin descriptions

2. Electrical Characteristics

2.1. Absolute Maximum Ratings

Parameters	Min	Max	Unit
Storage temperature	-40	+120	°C
Supply voltage VDD pin to ground	-0.5	+3.6	V
Voltage applied to inputs	-0.5	VDD+0.5	V
Short circuit duration, to GND or VDD		5	sec

Table 2 Absolute maximum ratings

2.2. Recommended Operating Conditions

Test conditions: VDD = 3 V

Parameters	Min	Typ	Max	Units
*Ambient Operating Temperature	-20		+85	°C
Supply Voltage for VDD_3V	2.4	3	3.6	V
Logical high input voltage (for DI type pins)	0.5XVDD_D			V
Logical low input voltage (for DI type pins)			0.2XvVDD_D	V

Table 3 Recommended operating conditions

2.3. DC Electrical Characteristics

Test conditions: T_A = 25°C, VDD = 3 V

Chip Mode	Condition	Min	Typ	Max	Unit
POWER DOWN	All circuit power is off, except the wake-up circuit.			1	uA
DEEP SLEEP	All circuit power off except voltage regulators.		2.5		uA
STANDBY I	All circuit power off except voltage regulators and partial 32MHz cloc		80		uA

ACTIVE: TX	@ 0 dBm output power @ 1Mbps		15		mA
ACTIVE: RX	@ 250 kbps		16		mA
	@ 1 Mbps		18		mA

2.4. Radio Frequency AC Characteristics

2.4.1. Receiver Radio Frequency AC Characteristics

Test conditions: T_A = 25°C, VDD = 3 V, LO frequency=2.445 GHz

Parameters	Condition	Min	Typ	Max	Unit
RF input frequency		2.405		2.480	GHz
RF sensitivity	At antenna input with O-QPSK signal (PER <= 0.1%) Normal mode (250 Kbps) Turbo mode (1 Mbps)		-90 -80		dBm dBm
Maximum RF input			+5		dBm
Adjacent channel rejection	+/-5 MHz @ 250kbps (-82dBm + 20 dB = -62dBm)		20 -62		dBc dBm
LO leakage	Measured at the balun matching the network with the input frequency at 2.4 ~ 2.5 GHz		-60		dBm
Noise figure (Including matching)			8		dB
Alternative channel rejection	@+/-10 MHz (-82dBm + 40 dB = -42dBm)		40 -42		dBc dBm
Total RX current			18		mA

2.4.2. Transmitter Radio Frequency Characteristics

Test conditions: TA = 25°C, VDD = 3 V, LO frequency=2.445 GHz

Parameters	Condition	Min	Typ	Max	Unit
RF carrier frequency		2.405		2.480	GHz
Maximum RF output power		-3	0		dBm
RF output power Accuracy				±4	dBm
RF output power control range			36		dB
Carrier suppression			-30		dBc

TX spectrum mask for O-QPSK signal	Offset frequency > 3.5 MHz, at 0 dBm output power		-30		dBm
TX EVM			30%		
Total TX current	At 0 dBm output power		15		mA

2.4.1. Synthesizer Characteristics

Test conditions: TA = 25°C, VDD = 3 V, LO frequency=2.445 GHz

Parameters	Condition	Min	Typ	Max	Unit
PLL Stable Time			130		us
PLL Programming resolution			1		MHz

2.4.1. Crystal Parameter Specifications

Parameter	Min	Typ	Max	Unit
Crystal Frequency		32		MHz
Frequency Offset	-40		40	ppm

2.5. Peripheral Characteristics

UM2460 has slave mode SPI interfaces. They can be used by the host MCU to access UM2460 registers and FIFOs. The 4-wire SPI (SEN, SCLK, SI, SO) provides a high speed interface up to 5MHz on SCLK.

2.6. Power-on and Reset Characteristics

UM2460 has built-in power-on reset (POR) circuit which automatically resets all digital registers when power is turned on. The 32MHz oscillator circuit starts to lock to the right clock frequency after power-on. The whole process takes 1ms for clock circuit to become stable and complete the power-on reset. It is highly recommended that the user waits at least 1ms before starting to access UM2460.

For external hardware reset (warm start), external reset pin RESETn is internally pulled-high. UM2460 will be in reset state after RESETn is released from low state.

2.7. Crystal Parameter Specifications

The clock system of UM2460 can be separated to two parts, one is called main clock and the other is called sleep clock. Among all, the 32MHz clock utilizes external crystals to generate the oscillations. The associated pins are XTAL_P, XTAL_N. The frequency variation allowed for 32MHz crystal is from -40ppm to +40ppm. The sleep clock is generated by internal ring oscillator.

3. Functional Description

3.1. Memory Block

The memory block (including registers and FIFOs) of UM2460 is implemented by SRAM. It is composed of registers and FIFOs as shown below:

Registers

- Short register (6-bit short addressing mode register, total 64 bytes)
- Long register (10-bit long addressing mode register, total 128 bytes)

FIFOs

- TX FIFO – TXFIFO (64 bytes)
- RX FIFO – RXFIFO0 (64 bytes), RXFIFO1 (64 bytes)

Registers provide control bits and status flags for UM2460 operations, including transmission, reception, interrupt control, timer, MAC/baseband/RF parameter settings, etc. Short registers are accessed by short addressing mode with valid addresses ranging from 0x00 to 0x3F. Long registers are accessed by long addressing mode with valid addresses ranging from 0x200 to 0x27F.

FIFOs serve as the temporary data buffers for data transmission and reception. Each FIFO holds one packet only at a time. The transmission FIFO is called TXFIFO. TXFIFO is composed of 64-byte FIFO. The receiving FIFO is called RXFIFO. RXFIFO is composed of one 64-byte FIFO.

3.1.1. Registers

Short Registers

Short registers are accessed by short addressing mode with valid addresses ranging from 0x00 to 0x3F. Together with long registers, they provide control bits and status flags for UM2460 operations, including transmission, reception, interrupt control, timer, MAC/baseband/RF parameter settings, etc.

Short registers are accessed faster than long registers.

Long Registers

Long registers are accessed by long addressing mode with valid addresses ranging from 0x200 to 0x27F. Together with short registers, they provide control bits and status flags for UM2460 operations, including transmission, reception, interrupt control, timer, MAC/baseband/RF parameter settings, etc.

Long registers are accessed slower than short registers.

3.1.2. FIFOs

RXFIFO

RXFIFO is composed of two 64-byte FIFO memory (RXFIFO0 and RXFIFO1) to store the incoming packet. Each of them is designed to store one packet at a time. Section 3.7 details the two RX FIFO behavior.

3.1.3. Memory Space

UM2460 memory is divided into two addressing spaces. One is the short addressing space; the other is the long addressing space. The short addressing space is from 0x00 to 0x3F. The long addressing space is from 0x001 to 0x33F. The first byte 0x000 of long addressing space is un-used.

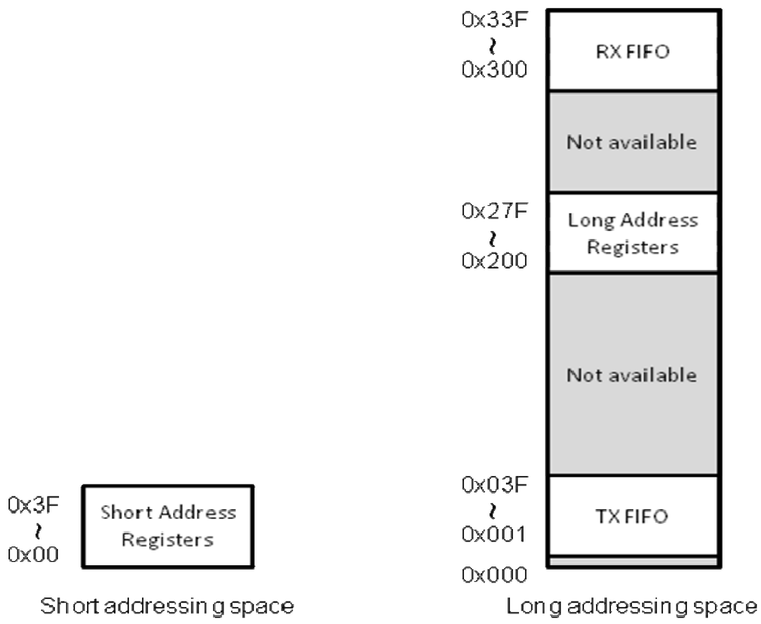

Fig. 1 Memory space diagram

3.2. TX Packet format

Figure 2 shows the TX packet format which contains a length field, address field, frame control field and payload field. TX packet is stored in TX FIFO. The address of TX FIFO is from 0x001 to 0x03f.

The first one byte of TX packet format specifies the length of the address field, frame control field and the payload field. The valid TX length can be from 5 to 61 bytes.

The address field should contain the broadcast address (0xffff-ffff) or RX destination address which can be configured by SREG05, SREG06, SREG07 and SREG08 as Figure 3.

The LSB of frame control field is used for acknowledgment request which specifies whether an acknowledgment is required from the recipient device. If the bit is set to 1, the recipient device shall send an acknowledgment frame after determining that the frame is valid. If the bit is set to 0, the recipient device shall not send an acknowledgment frame after determining that the frame is valid.

The bit 1 of frame control field is used for frame type. If the bit is set to 1, the frame type is acknowledgment frame. If the bit is set to 0, the frame type is data frame.

The payload is the user defined content of the packet. It has variable length from 0 to 56 bytes.

Normally, TXMAC will generate the CRC data and append the CRC data with the end of the packet automatically. If setting the bit 7 of SREG00 to 1, no CRC data appended.

Name	Size	Description
LEN	1 byte	Data length (5~61)
ADDR	4 bytes	Destination address
FC	1 byte	Frame control, FC[1] : ACK frame FC[0] : ACK request
PL	0~56	Data

Fig. 2 TX packet format

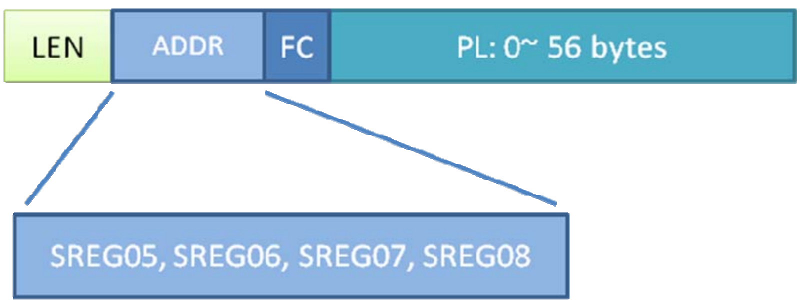

Fig. 3 address registers

3.3. RX Packet format

Figure 4 shows the RX packet format which contains a length field, address field, frame control field, payload field and CRC field. RX packet is stored in RX FIFO. The address space of RX FIFO is from 0x300 to 0x33f. RX packet is including CRC field. The length field indicates the length of the address field, frame control field, the payload field and CRC field. The valid RX length can be from 7 to 63 bytes.

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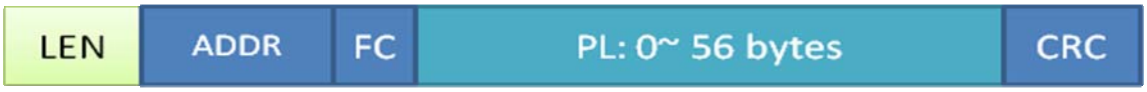
The CRC field is 2 bytes in length and contains a 16 bit ITU-T CRC. The CRC is calculated over the address field, frame control field and the payload.

The CRC is calculated using the following standard generator polynomial of degree 16:

$X^{16} + X^{12} + X^5 + 1$



Name	Size	Description
LEN	1 byte	Data length (7~63) (=TX_length+2)
ADDR	4 bytes	Destination address
FC	1 byte	Frame control, FC[1] : ACK frame FC[0] : ACK request
PL	0~56	Data
CRC	2 bytes	CRC

Fig. 4 RX packet format

3.4. Acknowledgment format

Figure 5 shows the acknowledgment packet format which contains frame control field (FC[1]=1), information field and CRC field. The length field indicates the length of frame control field, the information field and CRC field. The length of acknowledgment frame is always 5 bytes. The information field is the user defined content which can be configured by SREG03 and SREG04. The CRC field is 2 octets and contains a 16-bit ITU-T CRC. The CRC is calculated over the CRC of received frame, frame control field and information field.

Name	Size	Description
LEN	1 byte	Frame length(=5)
FC	1 byte	Frame control, FC[1] =1: ACK frame
INF	2 bytes	User information (stored in SREG03, SREG04)
CRC	2 bytes	CRC

Fig. 5 Acknowledgment frame format

3.5. Hardware Acknowledgement

The request for remote acknowledgement of a transmitted packet can be enabled or disabled by setting its frame control field.

Bit 0 of the “Frame control” field indicates whether the frame needs an acknowledgement or not. Users should prepare the header information correctly and write it into TXFIFO. If “Ack request” bit-field is set to “1”, the receiver of this packet is required to send the ACK packet back. If the ACK frame from the remote receiver is not received, the transmitter should send the packet again and again till the maximum retransmission times is reached. SREG1b[7:4] is the maximum retransmission times.

SREG1B: TXNMTRIG
Offset: 0x1b

Bits	Name	Description	Reset Value	R/W
7-4	**txnretryn**	Max. retry times of the most recent TXNFIFO transmission.	0	R/W
3	**Reserved**			
2	**ackreq**	TX packet in TXNFIFO needs ACK response.	0	R/W
1	**Reserved**			
0	**txstart**	Trigger TXMAC to send the packet in TXFIFO.	0	WT

UM2460 has built-in circuit to facilitate the acknowledge protocol automatically. For transmitting side, UM2460 supports auto-retransmission. For receiving side, UM2460 supports auto-acknowledgement. Each of the two sides needs to set corresponding registers correctly to utilize the functions.

Auto-retransmission on TX Side

For TXMAC to retransmit a packet automatically when ACK is not received, “TXNACKREQ” (SREG0x1B[2] for TX normal FIFO) is required to be set to “1”.

Auto-acknowledgement on RX Side

RXMAC of UM2460 will automatically reply ACK packet by default if “Ack request” in the frame header is set to “1”. This auto-acknowledgement feature can be disabled by setting NOACKRSP (SREG0x00[5]) to “1”.

3.6. Primary TX Operations

You activate primary TX mode by setting SREG0d[2] to “1”. After changing the SREG0d[2] value, you have to reset RF and let RF state machine go to primary TX mode correctly. If primary TX mode is enable, the UM2460 will enter the sleep mode after any packet transmits. If primary TX mode is not enable, the UM2460 will enter the RX mode after any packet transmits. If transmit packet in TXFIFO needs ACK response and primary TX mode enable, the UM2460 will enter the sleep mode after ACK frame received. If no ACK frame received, the UM2460 will not enter the sleep mode until the max time to wait for an acknowledgment frame by setting SREG12[6:0].

SREG12: ACKTMOUT
Offset: 0x12

Bits	Name	Description	Reset Value	R/W

7	**Reserved**			
6-0	**mawd**	The maximum number of symbols to wait for an acknowledgment frame to arrive following a transmitted data frame.	0x39	R/W

3.7. Two RX FIFOs

RXMAC block will do the CRC checking, parsing the received frame type and do the address recognition before storing the received frame in the RXFIFO which contains two 64-byte dual ports FIFOs, RXFIFO0 and RXFIFO1. The received frame will be stored in RXFIFO one packet at a time. A byte of length will be extracted from PHY header and be appended in front of the MAC frame. This facilitate the host MCU to decode the frame correctly with correct frame length information.

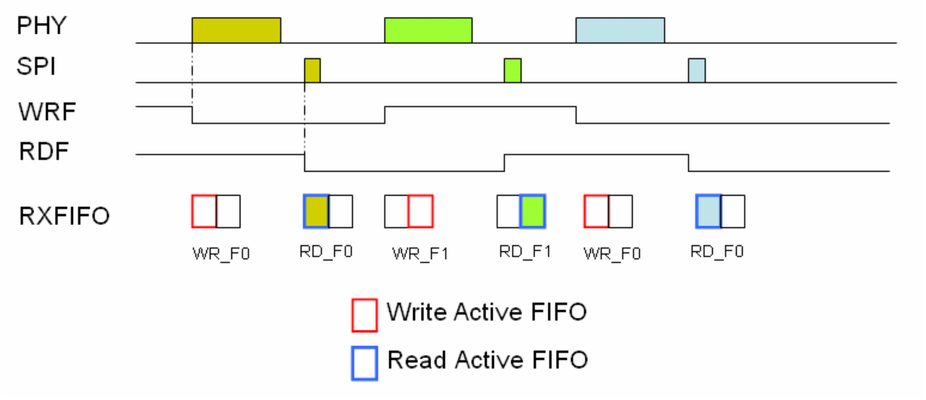
The behaviors of RXFIFO follow a certain rule: When a received packet is not filtered or dropped out, a received interrupt/status will be issued to host MCU. The interrupt is a read-to-clear type to save the host MCU operation time. However, the contents of RXFIFO can be flushed only by the following three ways: (1) the host MCU reads the first byte of the packet, (2) the host issues an RX flush, and (3) the software reset. Please note that once the first byte of RXFIFO is read, RXFIFO is ready to receive the next packet and all the data will be overwritten. So it is suggested that the host MCU reads back all data without any interrupt or jumping to other process.

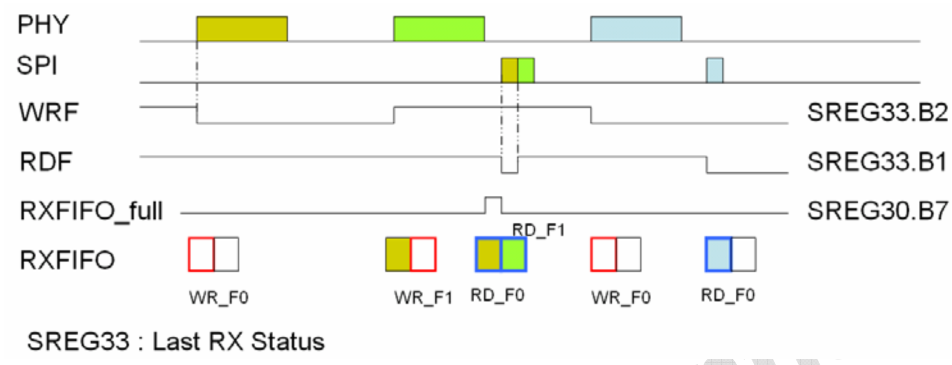
If the address is correct address or broadcast address, RXMAC will issue RX interrupt to host device to indicate a valid packet is received. UM2460 RXMAC also supports promiscuous mode and error mode. Promiscuous mode is supported to receive all FCS-ok packets. Error mode is supported to receive all packets that successfully correlate with the PHY level preamble and delimiter. Under these two conditions, RXMAC issues RX interrupt, too.

UM2460 RXMAC supports automatic ACK reply. If and only if the five conditions mentioned above are met, and AckReq bit in frame-control field is set, an ACK packet will be sent by TXMAC automatically in the meantime.

Two Ping c FIFOs, RXFIFO0 and RXFIFO1, are mapped to the long address memory space from 0x300 to 0x38F. UM2460 supports Ping-Pong receiving process utilizing the two FIFOs. This receiving process is called Ping-pong RX mode. Each FIFO has 64-byte memory with two ports. The bit-field SREG0x34[0] can enable the Ping-Pong FIFO mode, which is disabled by default. RXMAC automatically switches between RXFIFO0 and RXFIFO1 whenever a new packet comes.

An RX interrupt will be issued to the host MCU after the complete frame is stored into RXFIFO. For Ping-pong RX mode, when the host MCU read the first byte of the long address memory LREG0x300, RXMAC will change the read flag SREG0x34[1] automatically. For manually controlled RX operation, if the value of the read flag SREG0x34[1] is 0, RXFIFO0 will be read. Otherwise if the value of the read flag SREG0x34[1] is 1, RXFIFO1 will be read.





In the above diagram, the current status of each frame is represented as the register SREG0x30. The bit 7 of the register SREG0x30 means RXFIFO full indicating the two RXFIFOs are occupied. If the host MCU can not read the RXFIFO in time, the value of SREG0x30[7] will be set to 1, and the value of SREG0x30[6:1] will be replaced. Then the last status of the RX frame will be kept in SREG0x33. Once the host MCU read the RXFIFO, the value of the SREG0x30[7] will be set to 0 automatically.

SREG30: RXSR
Offset: 0x30

Bits	Name	Description	Reset Value	R/W
7	**RXFFFull**	RX FIFO full	0	R
6	**WrFF1**	0: write data to RX FIFO 0 1: write data to RX FIFO 1	0	R
5	**Reserved**			
4	**RXFFOvfl**	RXFIFO overflow	0	R
3	**RXCrcErr**	RX CRC error flag, update when received each packet	0	R
2-0	**Reserved**			

SREG33: Last RX Status
Offset: 0x33

Bits	Name	Description	Reset Value	R/W
7	**LRXFFFull**	RX FIFO full status of last frame	0	R
6	**LWrFF1**	0: write last packet to RX FIFO 0 1: write last packet to RX FIFO 1	0	R
5	**Reserved**			
4	**LRXFFOvfl**	RXFIFO overflow status of last frame	0	R
3	**LRXCrcErr**	RX CRC error flag of last frame	0	R
2-0	**Reserved**			

SREG34: SPI Mode & RX two FIFO
Offset: 0x34

Bits	Name	Description	Reset Value	R/W
7-6	**SPI_Mode**	SPI Mode: 00: normal 01: PLUS	00	R/W
5	**BatteryInd**	Battery low indicator	0	R
4-3	**Reserved**			
2	**WrFF1**	0: write data to RX FIFO 0 1: write data to RX FIFO 1	0	R
1	**RdFF1**	0: read data from RX FIFO 0 1: read data from RX FIFO 1	0	R
0	**RXFIFO2**	Set RX two FIFO mode active	0	R/W

3.8. Power Management

Almost all wireless sensor network applications require low-power consumptions to lengthen the battery life. Typical power consumption of a battery-powered device is such that the deployed device could operate over years without replacing its battery. The UM2460 achieves low active current consumption of both the digital and the RF/analog circuits by controlling the supply voltage and using low-power architecture. The sleep current for UM2460 can be as low as 1uA only.

For ultra low-power operation, a Power-down mode is available which consumes around 1uA while UM2460 is powered down. All data stored in registers and FIFOs will be lost under Power-down mode. However in this mode, UM2460 is able to wake up by an external wake-up signal.

3.8.1. External bypass capacitors

The table below lists the recommended external bypass capacitors for each pin of the UM2460. For the three power supply pins (pin 1, pin26 and pin 30), they need an extra bypass capacitor in parallel for decoupling purpose while the rest of the power supply pins require only one bypass capacitor. The path length between the bypass capacitors to each pin should be made as short as possible.

Pin	Symbol	Bypass Capacitor 1	Bypass Capacitor 2
1	VDD_RF1	47 pF	10 nF
7	VDD_RF2	47 pF	
24	VDD_D	10 nF	
26	VDD_3V	47 pF	10 nF
29	VDD_GR	100 nF	
30	VDD_BG	47 pF	10 nF
31	VDD_A	47 pF	
35	VDD_PLL	47 pF	
37	VDD_CP	10 nF	
38	VDD_VCO	1 uF	

Table 4 Recommended external bypass capacitors

3.8.2. Voltage Regulators

UM2460 has an internal DC-DC regulator to generate core voltage(2.2V to 1.8V) efficiently to each power pin through external connections. The transfer efficiency is largely increased and current consumption can be reduced. The external global power source should be applied to Pin 26(VDD_3V) which is the input node of DC-DC regulator. The DC-DC regulator generates the regulated voltage power and exports the voltage to Pin 27(VDD_2V2) with typical value of 2.2V which can be adjusted manually. This 2.2V voltage supply can be used as a power supply to bias the rest of the supply voltage pins of UM2460.

For each of the supply voltage pin of UM2460, there is an individual voltage regulator, which allows input voltage from 3.6V to 2.4V, connected right after the IO pad of the supply voltage pin. No external stabilizing capacitor is needed for each of the LDO voltage regulators. All these LDO regulators generate 2.4V output to internal circuit nets. User can connect all these supply voltage pins to Pin 10 thereby biasing UM2460 with DC-DC regulator, or bias them with other external power sources with the internal DC-DC regulator turned-off optionally. The DC-DC regulator can be turned off by setting LREG0x250[2] to ‘0’.

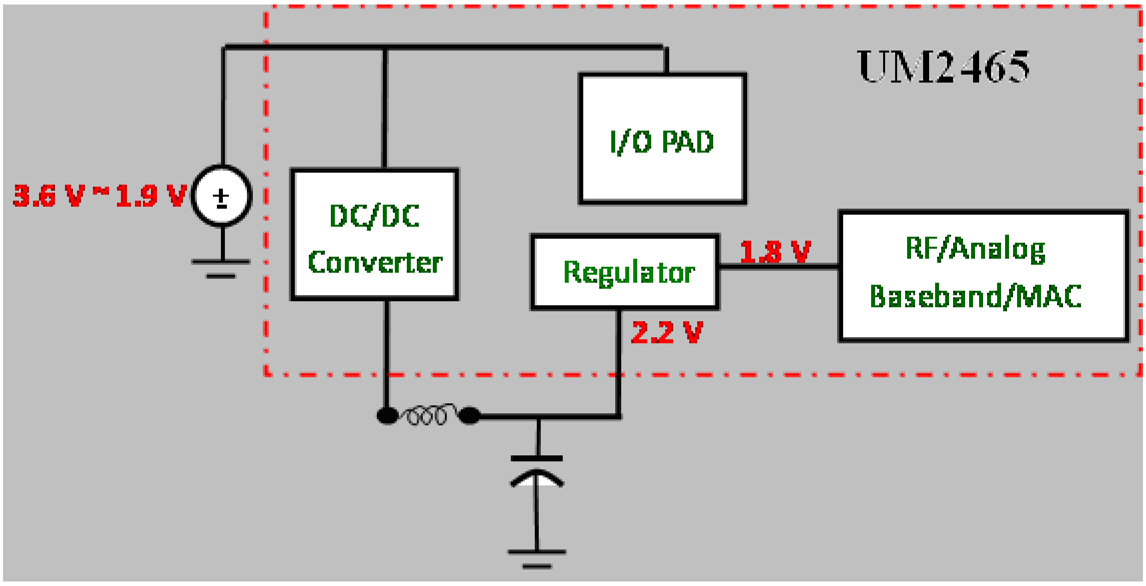


Fig. 7 Block diagram of voltage regulators

3.8.3. Battery Monitor

UM2460 provides a function to monitor the system supplied voltage. A 4-bit voltage threshold can be configured so that when the supplied voltage is lower than the threshold, the system will be notified. For battery monitor function, please refer to Section 4.5.

3.8.4. Power Modes

The power modes of UM2460 are classified into the following five modes:

- ACTIVE : Fully turned on with two sub-modes designated as TX-ACTIVE and RX-ACTIVE
- STANDBY : RF/MAC/BB shutdown with sleep and 32MHz clocks remain active
- DEEP_SLEEP : All power is shutdown except the power to the digital circuits and DC-DC regulator. register and FIFO data are retained.
- POWER_DOWN : All power is shutdown. Register and FIFO data are not retained, an external wake-up signal can wake up UM2460.

3.8.5. Sleep and Wake-up

3.8.5.1. Wake Count setting

WAKECNT is used for the 32 MHz clock of the UZ2460 to recover from sleep mode.

WAKECNT = {SREG0x36[7:0], SREG0x35[6:0]}

Step 1.
Calibrate the sleep clock frequency:

- The UZ2460 uses a 16 MHz system clock to calibrate the sleep clock frequency. To do the calibration, set LREG0x20B[4] to the value '1', and then keep polling LREG0x20B until the value of LREG0x20B[7] becomes '1'. After the value of LREG0x20B[7] becomes '1', LREG0x20B[3:0], LREG0x20A, LREG0x209 form a 20-bit value C. Then the period of the sleep clock (Psleepclock) can be calculated by the following equation,

$$P_{\text{sleepclock}} = \frac{62.5 \times C}{16}(ns)$$

Step 2

Configure the recovery time of the 32 MHz clock by setting {SREG0x36[4:3], SREG0x35[6:0]} to max value 500 us.
For example, the period of the sleep clock (Pslowclock) is 10 us.
Please set {SREG0x36[4:3], SREG0x35[6:0]} to 0x32

3.8.5.2. *Sleep*

UM2460 has two sleep methods:

- ***Sleep Acknowledgement (SLPACK, SREG0x35[7])***

The SLPACK (SREG0x35[7]) should be issued by host MCU to initiate the sleep process without using sleep clock.

- Trigger sleep mode using sleep clock

The StartCnt (LREG0x229[7]) should be set to "1" to trigger to enter sleep mode using sleep counter.

3.8.5.3. *Wake-up*

- Immediate wake-up
  - UM2460 can be awakened externally by WAKE (pin 15). SREG0x0D[5] enables the WAKE pin. SREG0x0D[6] controls the polarity of the WAKE pin. Be sure to set SREG0x22[7] to "1" to enable the Immediate wake-up mode before sleep.
  - UM2460 can also be awakened by setting SREG0x22[6] to "1" and back to "0", the immediate wake-up mode is then entered.

3.9. Analog Circuits

3.9.1. PLL Frequency Synthesizer

The loop filers of the Phase-Lock Loop (PLL) in the frequency synthesizer are integrated into UM2460 except one external capacitor which should be connected between pin 48 and the ground. The board layout around pin 48 should be carefully designed to avoid EMI (electro magnetic interference) in order to keep the PLL stable.

The recommended value of this external capacitor is 100 pF.

3.9.2. Internal Sleep Clock Oscillator for Sleep Clock

There is a free-running sleep clock oscillator integrated into UM2460. No external component is needed for the operation of this sleep clock. User has to calibrate this sleep clock before an application starts.

3.10. Peripherals

3.10.1. SPI Interface

The SPI module provides slave roles SPI mode 0 interface to read/write the control registers and FIFO of UM2460. The features are as below:

- A simple 4-wire SPI-compatible interface (pins SCLK, SEN, SI and SO) where UM2460 is the slave.
- Most significant bit (MSB) of all address and data transfer on the SPI interface is done first

SPI Characteristics

Parameter	Symbol	Min	Max	Units	Conditions
SCLK, clock frequency	F_{SCLK}		<5	MHz	
SCLK low pulse duration	t_{CL}	50		ns	The minimum time SCLK must be low.
SCLK high pulse duration	t_{CH}	50		ns	The minimum time SCLK must be high.
SEN setup time	t_{SP}	50		ns	The minimum time SEN must be low before the first positive edge of SCLK.
SEN hold time	t_{NS}	>50		ns	The minimum time SEN must be held low after the last negative edge of SCLK
MOSI setup	t_{SD}	25		ns	The minimum time data must be ready at MOSI, before the positive edge of SCLK
MOSI hold time	t_{HD}	25		ns	The minimum time data must be held at MOSI, after the positive edge of SCLK
Rise time	t_{RISE}		25	ns	The maximum rise time for SCLK and SEN.
Fall time	t_{FALL}		25	ns	The maximum fall time for SCLK and SEN.

Table 5 SPI characteristics

SPI Frame Format

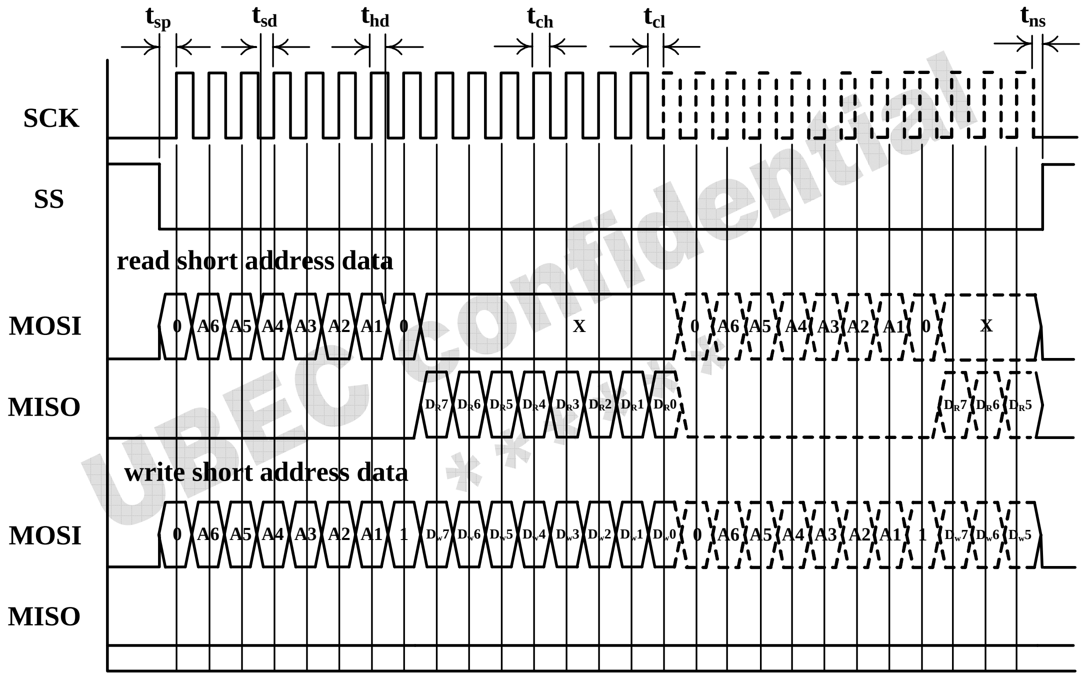
Short addressing format

bit [7]	bit [6:1]	bit [0]
Short Address 0	0x3F ~ 0x00	Read / Write 0 1

Long addressing format

bit [11]	bit [10:1]	bit [0]
Long Address 1	0x27F ~ 0x200	Read / Write 0 1

SPI Timing Diagrams



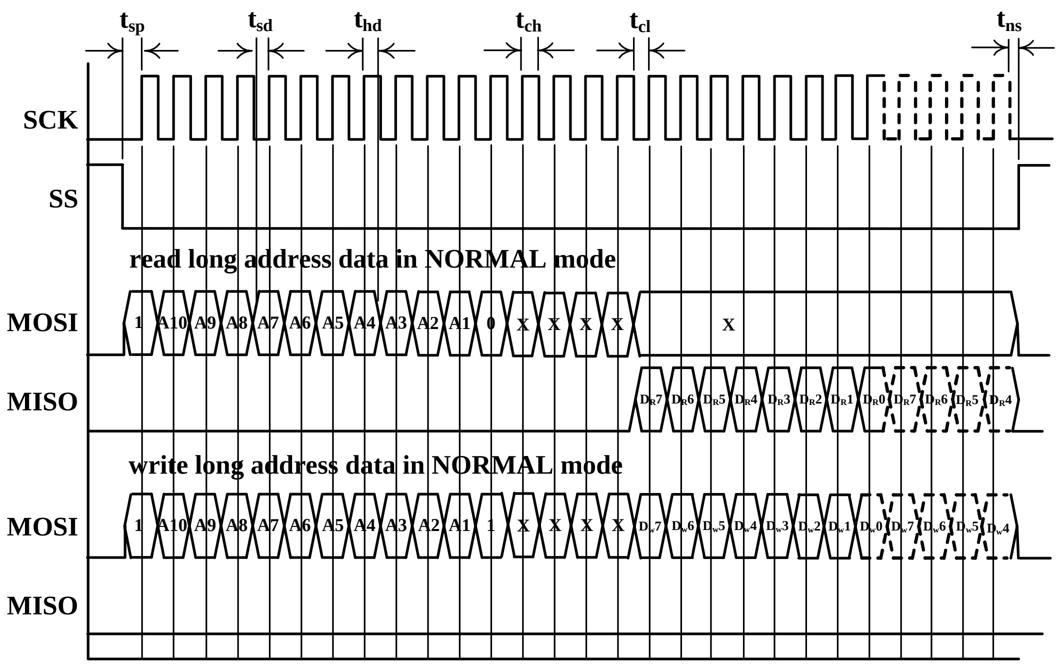


Fig. 10 Timing diagram and specification

3.10.2. GPIO

UM2460 has 3 digital GPIOs. Each GPIO can be configured as input or output respectively. When being configured as an output pad, the driving capability is 4mA for GPIO0 and 1mA for GPIO1, and GPIO2.

To benefit long range applications, GPIO1 and GPIO2 can be configured to control external PA and LNA, switch according to the current RF state automatically. Please refer to Section 4.1.2 for external PA/LNA application configuration.

3.10.3. Interrupt Signal

UM2460 provides an output interrupt pin (INT) and the polarity of interrupt signal is selectable. UM2460 issues interrupts to host MCU on two possible events. If any event happens, UM2460 sets the corresponding status bit in SREG0x31. And if the corresponding interrupt mask in SREG0x32 is clear (i.e. equals “0”), interrupt will be issued. If it is set (masked, blocked), interrupt will not be issued, but the status is still present. Interrupt is a read-to-clear operation. Whenever SREG0x31 is read, interrupt and status are cleared. This is beneficial to increase the performance.

UM2460 issues interrupt according to the following eight events which are described as below:

RX OK interrupt (RXIF)

This interrupt is issued when an available packet is received in RXFIFO. An available packet means that it passes RXMAC filter, which includes FCS check, PANID/address filtering, packet type or promiscuous/error modes.

TX normal FIFO release interrupt (TXNIF)

This interrupt is issued when a packet in normal FIFO is triggered and sent successfully or is triggered and the retransmission is timed out. SREG0x24[0] indicates the TX normal FIFO release status.

The interrupt events are indicated in SREG31. The register is “read-to-clear”. Each of the eight interrupts can be masked by setting INTMSK(SREG0x32).

4. Application Guide

Some typical applications are described in this chapter to help designer gains more understanding of the capability of UM2460.

4.1. Hardware Connection

4.1.1. Typical Application Connection Using SPI Interface

A typical connection using SPI interface is shown below. The MCU serves as master role and UM2460 serves as slave role.

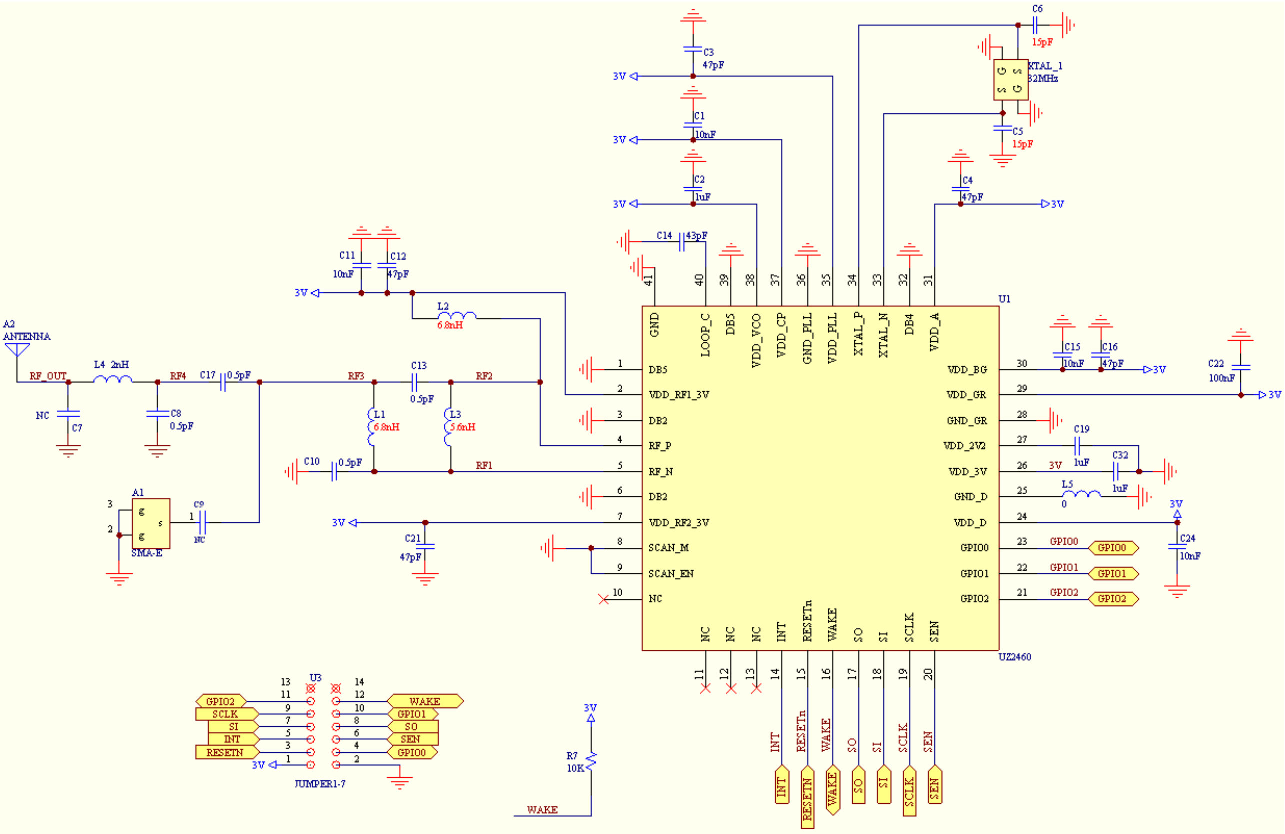


Fig. 11 Typical application circuit using SPI interface

4.1.2. Adding External Power Amplifier and Low Noise Amplifier

The following depicts the typical circuit diagram for external PA/LNA enabled design using UBEC UP2268. UP2268 integrates PA, LNA and RF switch.

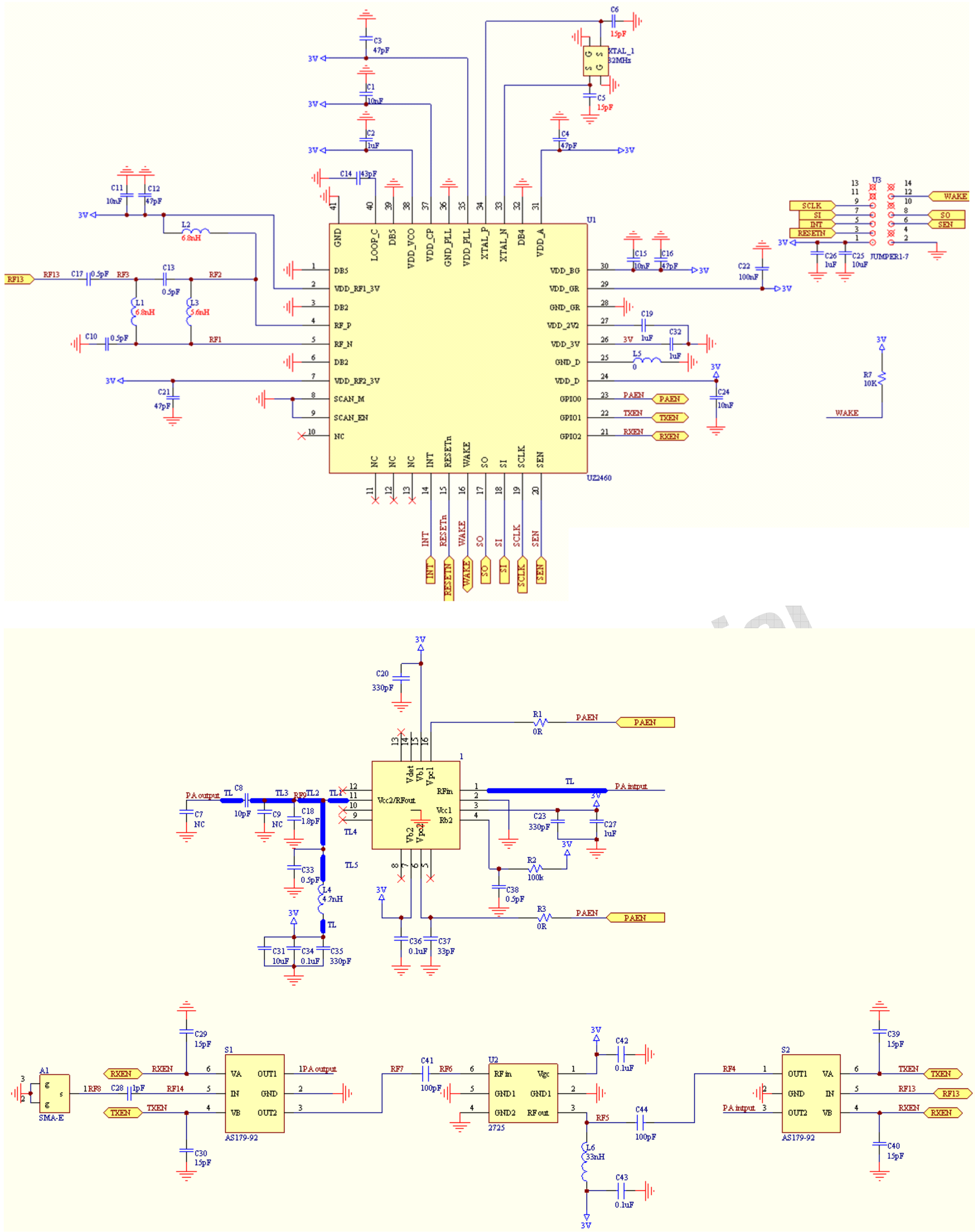


Fig. 12 Typical application circuit using external PA/LNA

4.1.3. External Power Amplifier Configuration

Set LREG0x22F to 0x0F to enable PA function. This register setting integrates the PA enable and RF Switch Control (TX branch, RX branch) by utilizing pin GPIO0, pin GPIO1, and pin GPIO2. If UM2460 is in TX mode, the pin GPIO0 (enable the external PA) and pin GPIO1 (TX branch enable) will be pulled HIGH, and the pin GPIO2 (RX branch enable) will be pulled LOW automatically. If UM2460 is in RX mode, the pin GPIO0 and pin GPIO1 will be pulled LOW, and the pin GPIO2 will be pulled HIGH automatically. That is, it automatically changes the status of the external PA and the TX/RX branch corresponding to the UM2460 TX or RX modes.

- TX mode: GPIO0 HIGH, GPIO1 HIGH, GPIO2 LOW
- RX mode: GPIO0 LOW, GPIO1 LOW, GPIO2 HIGH

LREG0x22F TESTMODE

LREG0x22F, TESTMODE							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit1	Bit 0
reserved					TESTMODE		
					R/W-000		

Bit 2-0 **TESTMODE**: UM2460 test mode using GPIO
0b101: Single-tone test mode.
0b111: GPIO0, GPIO1, GPIO2 are configured to control external PA, LNA or switch as discussed above.

4.2. Typical TX Operations

The TXMAC inside the UM2460 will automatically generate the preamble and Start-of-Frame delimiter fields when transmitting. Additionally, the TXMAC can generate any padding (if needed), and the CRC, if configured to do so. The host MCU must generate and write all other frame fields into the buffer memory for transmission described as followings.

4.2.1. Transmit Packet

To send a packet in TX FIFO, there are several steps to follow:

Step 1.
Fill in TX FIFO with packet to send. FCS is not necessary. The format is as follows

LEN	ADDR	FC	PL: 0~ 56 bytes

Table 6 TX FIFO format

Step 2.
Set ackreq (SREG0x1B[2]) if acknowledgement from the receiver is required. If Ackreq is set, UM2460 automatically retransmit the packet if there is no acknowledgement sent back. UM2460 will retransmit the packet till the Max trial times by SREG1b[7:6]..

Step 3.

Set the trigger bit (SREG0x1B[0]) to send packet. This bit will be automatically cleared. At this time, TXMAC will send the packet immediately.

Step 4.

Wait for the interrupt/status (SREG0x31[0])

Step 5.

If retransmit is required, check SREG0x24[0] to see if it is successful. “Successful” means that the ACK was received. If no retransmission is required, SREG0x31[0] indicates the packet is transmitted. Retransmission times can be read at SREG0x24[7:6].

SREG1B: TXNMTRIG
Offset: 0x1b

Bits	Name	Description	Reset Value	R/W
7-6	**txnretryn**	Max. retry times of the most recent TXNFIFO transmission.	0	R/W
5-3	**Reserved**			
2	**ackreq**	Transmit packet in TXFIFO needs ACK response.	0	R/W
1	**Reserved**			
0	**txstart**	Trigger TXMAC to send the packet in TXFIFO.	0	WT

SREG24: TXSR
Offset: 0x24

Bits	Name	Description	Reset Value	R/W
7-6	**txnretryn**	Retry times of the most recent TXNFIFO transmission.	0	R
5-1	**Reserved**			
0	**Txns**	Transmit successfully 0: ok, 1: fail(retry count exceed)	0	R

4.3. Typical RX Operations

UM2460 RX PHY filters all incoming signal and tracks the synchronization symbols. When preamble is found, the packet is demodulated and stored in the RXFIFO. At the same time, RXMAC starts calculating the frame FCS byte by byte and checking after having received a whole packet.
In the normal mode, RXMAC filters the MAC header and skips those packets not being sent to own addresses. If the packet is acceptable and needs an acknowledgement, RXMAC will inform TXMAC to send ACK packet automatically. In the promiscuous mode, the RXMAC receives all packets with correct FCS. In the error mode, the RXMAC will receive all packets.

4.4. Power Saving Mode Operations

The following provides an example setting for Power Saving Mode:

4.4.1. External Wake-up Mode

There are two modes for external wake-up. One is waken by external pin triggering , the other is waken by register triggering.

SREG22: TXBCNINTL

Offset: 0x22

Bits	Name	Description	Reset Value	R/W
7	**Imm_Wakeup**	Immediate wake-up mode enable 0: sync with slow clock 1: w/o slow clock	1	R/W
6	**Reg_Wakeup**	Register-based wake-up signal. Should be cleared to zero by SW.	0	R/W
5-0	**Reserved**			

Waken by External Pin

Configure pin 16 (WAKE),

- SREG0x22[7] = 0x0: set to external pin wake-up mode.
- SREG0x0D[6] = 0x1: set polarity of WAKE signal.
- SREG0x0D[5] = 0x0: turn on pad of the WAKE pin.

Waken by Register

Configure register.

- SREG0x22[7] = 0x1: set to register wake up mode.
- SREG0x22[6] = 0x1: wake up device.
-

4.5. Battery Monitor Operations

Step 1.

Set the battery monitor threshold at LREG0x205[7:4].

Step 2.

Enable the battery monitor by setting the LREG0x206[3] to "1".

Step 3.

Read the battery-low indicator at SREG0x34[5]. If this bit is set, it means that the supply voltage is lower than the threshold.

LREG0x205 RFCTRL5

LREG0x205, RFCTRL5							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit1	Bit 0
BATTH				reserved			
R/W-0b000				-			

Bit 7-4 **BATTH**: Battery monitor threshold value corresponding to voltage supply

0b1110 : 3.5V
0b1101 : 3.3V
0b1100 : 3.2V
0b1011 : 3.1V

0b1010 : 2.8V
0b1001 : 2.7V
0b1000 : 2.6V
0b0111 : 2.5V
Other : out of operation voltage

4.6. Initial Register Setting

After UM2460 is powered up, some registers need to be configured before the data transmission or reception. The procedure is described as below.

Step 1.

Set DC DC ON register.

Address mode	Addr	Setting Value(hex)	Notes
LREG	0x251	0xC0	
LREG	0x250	0x92	

When finish those settings, it need to delay 300 μSec to obtain the stable voltage.

//DELAY (300uS)   //Initial setting

Set the Initial registers according to the following table.

Address mode	Addr	Setting Value(hex)	Notes
SREG	0x21	0x55	
SREG	0x32	0x00	
SREG	0x35	0x7f	
SREG	0x38	0x80	80: 250K mode 81:1M Mode
SREG	0x3C	0x9C	
LREG	0x200	0x00	
LREG	0x201	0x03	
LREG	0x202	0x86	
LREG	0x203	0x00	
LREG	0x204	0x00	
LREG	0x205	0x00	
LREG	0x206	0x50	
LREG	0x207	0x00	00: 250K mode 10: 1M mode
LREG	0x208	0x08	Tx:08 Rx:C8
LREG	0x252	0x01	
LREG	0x253	0x00	
LREG	0x254	0x00	
LREG	0x255	0x00	
LREG	0x259	0x01	

LREG	0x273	0x00	
LREG	0x274	0xC8	
LREG	0x275	0x15	
LREG	0x276	0x01	
LREG	0x277	0x04	
SREG	0x2A	0x02	

4.7. Change channel

Step 1.

Set RF operation channel. UM2460 operates in the 2.4 GHz Industry, Scientific, and Medical (ISM) unlicensed band. The operating frequency is divided into 16 channels. If user wants to select any operating channel, RFCTL0(LREG0x200) should be configured as the following table.

Address mode	Addr	Setting Value(hex)
LREG	0x200	List as Table 8.

Channel Name	Frequency	Register Address	Set Value(hex)
Channel 11	2405 MHz	0x200	00
Channel 12	2410 MHz	0x200	10
Channel 13	2415 MHz	0x200	20
Channel 14	2420 MHz	0x200	30
Channel 15	2425 MHz	0x200	40
Channel 16	2430 MHz	0x200	50
Channel 17	2435 MHz	0x200	60
Channel 18	2440 MHz	0x200	70
Channel 19	2445 MHz	0x200	80
Channel 20	2450 MHz	0x200	90
Channel 21	2455 MHz	0x200	A0
Channel 22	2460 MHz	0x200	B0
Channel 23	2465 MHz	0x200	C0
Channel 24	2470 MHz	0x200	D0
Channel 25	2475 MHz	0x200	E0
Channel 26	2480 MHz	0x200	F0

Table 7 RF operation channel settings

Step 2.

Open the TX MAC gated clock by setting SREG26[1] to “1”.

Step 3.

After the operation channel is set, RF state machine should be reset by setting RFCTL(SREG0x36) to “0x04” and then setting RFCTL(SREG0x36) to “0x00”.

Address mode	Addr	Setting Value(hex)	Notes
SREG	0x36	0x04	
SREG	0x36	0x00	

By finishing all the above steps, a basic initialization procedure is done. This configuration procedure is valid for most of the application conditions.

Step 4.

After finished RF reset settings, it needs to delay 550 μSec to avoid to transmit ack frame during channel changed.

1Mbps mode : delay 300 μSec
250kbps mode : delay 550 μSec

Step 5.

To be disable the TX MAC gated clock by setting SREG26[1] to “0”.

Appendix A. TX Power Configuration

Default output power is 0dBm. Summation of “large” and “small” tuning decreases the output power.

LREG0x203 RFCTL3

Bits	Name	Description	Reset Value	R/W
7-3	TXG_B	TX back end stepped gain control (nominal value in dB) 00000: 0 dB; …… 11111: -9 dB	00000	R/W

Appendix B. Register Descriptions

Register Types

Register Type	Description
R/W	Read/Write register
WT	Write to trigger register, automatically cleared by hardware
RC	Read to clear register
R	Read-only register
R/W1C	Read/Write “1” to clear register

B.1 Short Registers (SREG0x00~SREG0x3F)

Revision History

Revision	Date	Description of Change
0.2	2009/6/3	Version 0.2 released.

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